Program 1 :

Docker file : Multi-stage builds using builder and production stages.

```
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$ ls
Dockerfile  node_modules  package.json  package-lock.json  src
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$ cat Dockerfile
#Base Image Name
FROM node:20-alpine AS builder


#Label names of the project like who is the maintainer project name and project version
LABEL maintainer="abhi@gmail.com"
LABEL project="Demo project"
LABEL version="1.0"

#The project working directory
WORKDIR /app

#copiyng all the package.json files so that can be used to install all the packages required to run the project
COPY package*.json ./


#npm ci-- meaning clean install
RUN npm ci

#Copy the rest of the files of the project to the working directory
COPY . .

RUN npm run build

# 2nd build state --- named as production
FROM node:20-alpine AS production

#Setting up the working directory
WORKDIR /app


#copying all the required files from the first build stage
COPY --from=builder /app/package*json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules



# Arguments that have been set so that can be imported in env varibales..
ARG NODE_ENV=production

#Environmental variables that required to run the project
ENV NODE_ENV=$NODE_ENV
ENV PORT=4000


#expose the port of the docker container
EXPOSE 4000


#in the terminal run the command node server.js to start the express application
CMD ["node","dist/server.js"]


# #Health check route monitoring every 30s so check whether the app is running and working fine
HEALTHCHECK --interval=30s --timeout=10s \
             CMD curl -f http://localhost:4000/health || exit 1
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$
```

Docker build logs

Command: docker build -t program1 .

```
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$ docker build -t program2 .
[+] Building 2.3s (14/14) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                            0.0s
 => => transferring dockerfile: 1.47kB                                                                          0.0s
 => [internal] load metadata for docker.io/library/node:20-alpine                                               1.9s
 => [internal] load .dockerignore                                                                               0.0s
 => => transferring context: 2B                                                                                 0.0s
 => [internal] load build context                                                                               0.2s
 => => transferring context: 44.92kB                                                                            0.1s
 => [builder 1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435  0.0s
 => CACHED [builder 2/6] WORKDIR /app                                                                           0.0s
 => CACHED [builder 3/6] COPY package*.json ./                                                                  0.0s
 => CACHED [builder 4/6] RUN npm ci                                                                             0.0s
 => CACHED [builder 5/6] COPY . .                                                                               0.0s
 => CACHED [builder 6/6] RUN npm run build                                                                      0.0s
 => CACHED [production 3/5] COPY --from=builder /app/package*json ./                                            0.0s
 => CACHED [production 4/5] COPY --from=builder /app/dist ./dist                                                0.0s
 => CACHED [production 5/5] COPY --from=builder /app/node_modules ./node_modules                                0.0s
 => exporting to image                                                                                          0.1s
 => => exporting layers                                                                                         0.0s
 => => writing image sha256:40e8906e4002daedd742a58ffeb07471a97f3cd8e7387e090336d375ec3d15ca                    0.0s
 => => naming to docker.io/library/program2                                                                     0.0s
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$ docker run -d -p 4000:4000 program2
a1a14fbb7d08cdba7c48ec71fc8efbdb62695644cc5ec953cc9181508f78b03d
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program2$
```

Running the docker built image in a container and mapping the port

command : docker run -d -p 3000:3000 program1